MiSeq i100 turbo recipe for faster run time

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Sequencing workflows are made up of individual modules

Collection and Transport

Pre-processing and Extraction

Library Prep

Sequencing

Analysis

Adjusting individual modules improves turnaround time

MiSeq i100
Turbo Recipe

Collection and Transport

Pre-processing and Extraction

Library Prep

Sequencing

Standard **7h30m**

Turbo **2h30m**

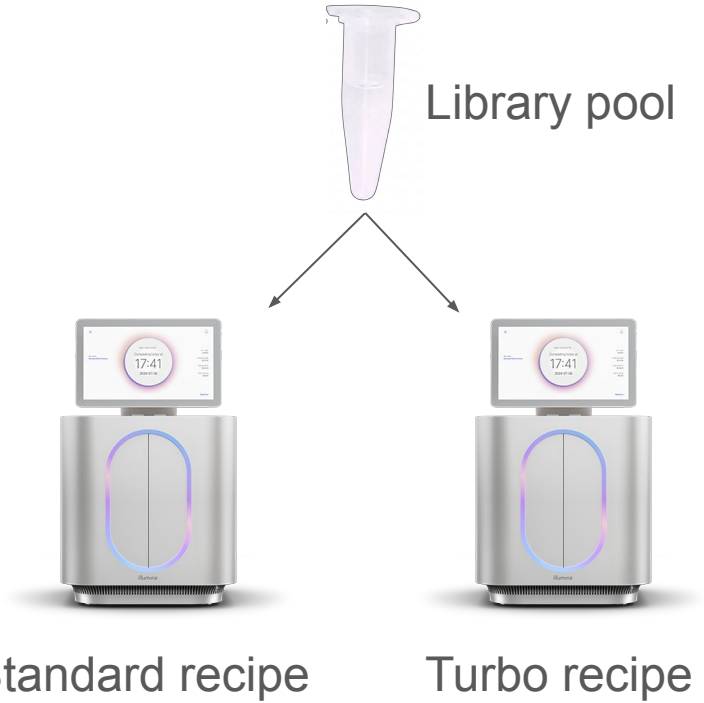
Analysis



Experimental Design

One identical library pool was run on MiSeq i100 twice:

- Standard recipe (~7h)
- Turbo recipe (~3h)



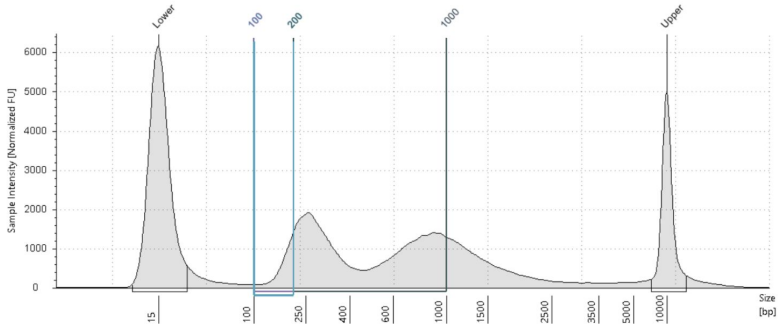
Setup

Samples

- Lung tissue biopsy
- 1x Shield neg ctrl (for lung tissue)
- Plasma
- 2x Shield neg ctrl (for plasma)

Library preparation

- Illumina Rapid RNA prep



Region Table

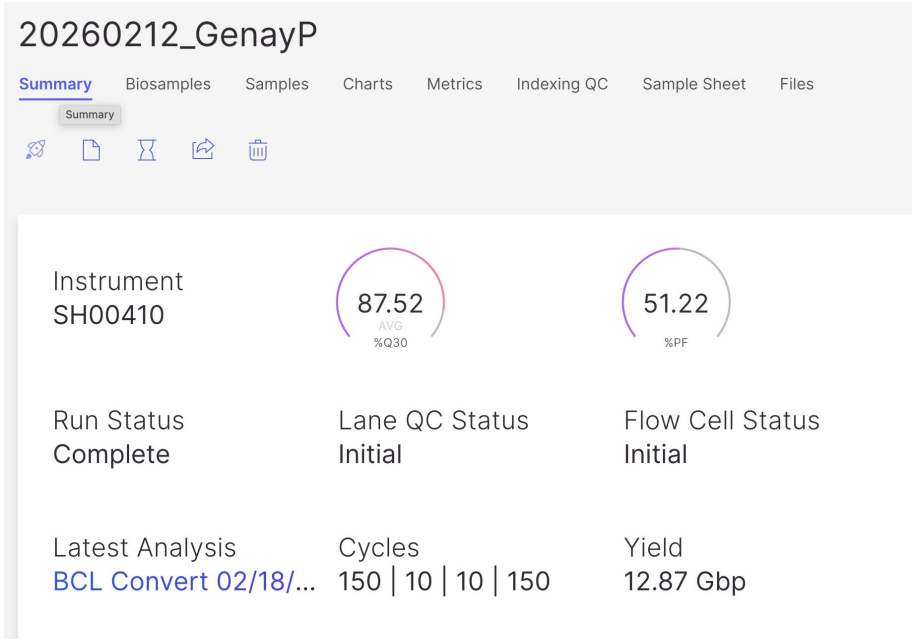
From [bp]	To [bp]	Average Size [bp]	Conc. [ng/µl]	Region Molarity [nmol/l]	%of Total	Region Comment	Color
100	200	172	1.12	10.4	6.46		
100	1000	487	11.5	49.8	66.05		
200	1000	521	10.5	40.2	60.13		

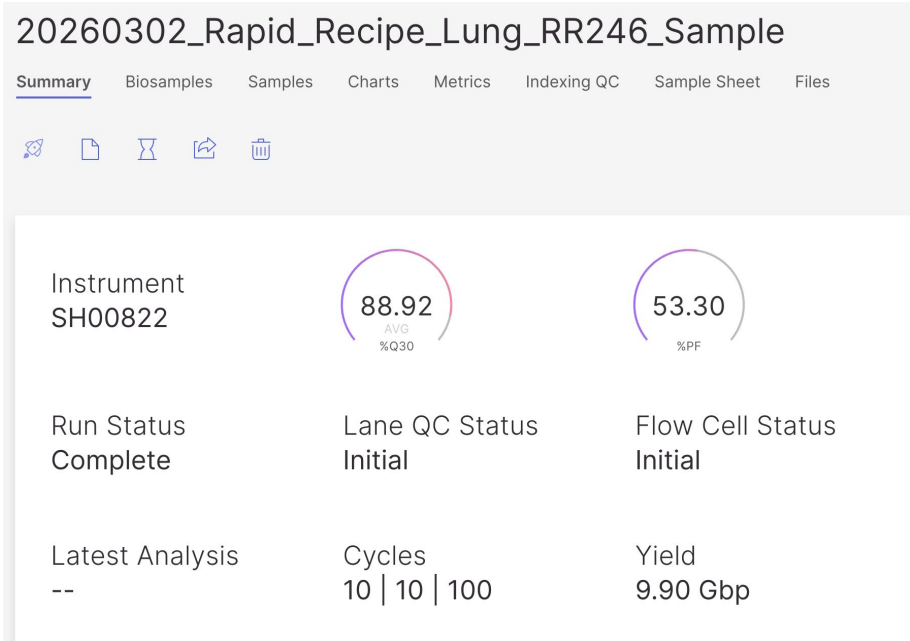
*Note: these libraries were over amplified

Summary of MiSeq i100 turbo recipe vs standard

	Standard	**Turbo recipe**
Recipe	Standard	Turbo
Run time	7h10m	3h29m
Kit	50M	100M
Clusters PF	40.5M	84.3M
Read length	2x150	1x100

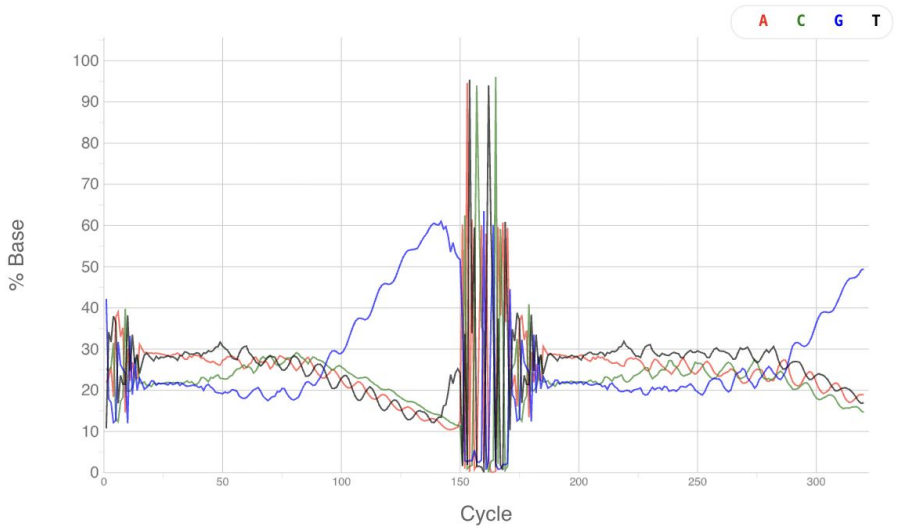
Seq run summary stats are similar; slightly higher for turbo



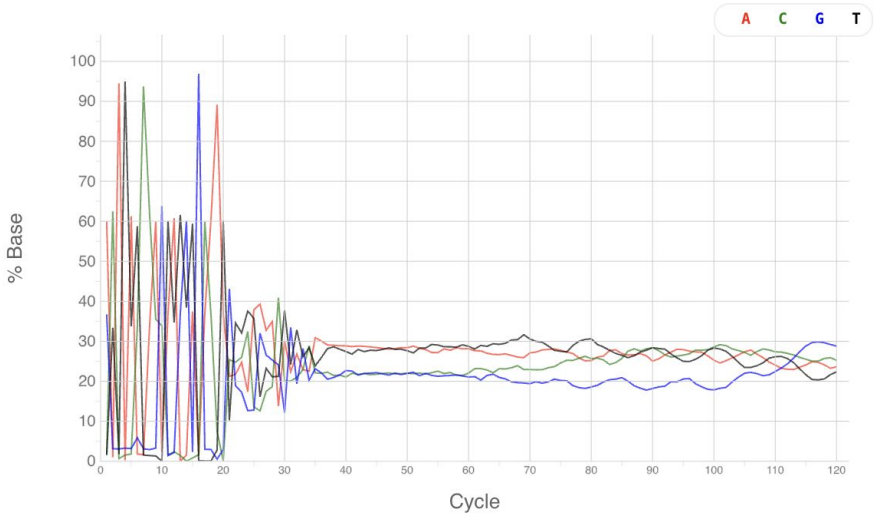


% base by cycle - inserts are < 150bp

Standard 2x150

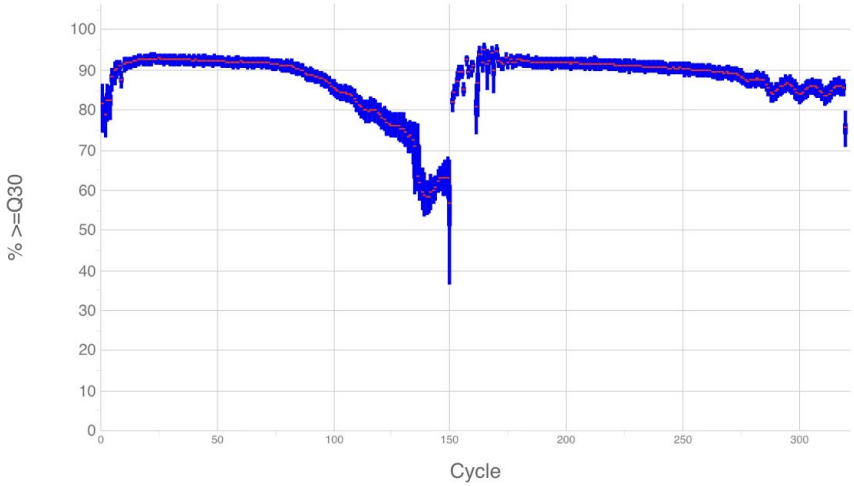


Turbo 1x100

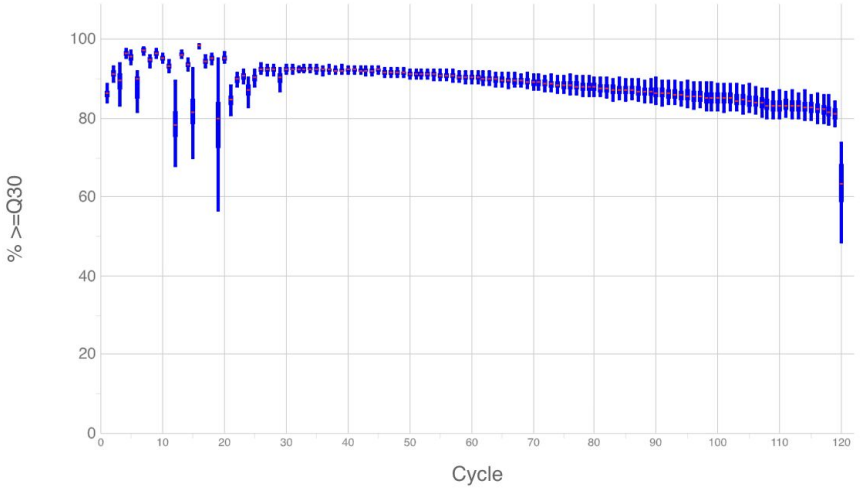


% >Q30 by cycle

Standard 2x150



Turbo 1x100



Turbo run looks good. Drop in standard run driven by short insert lengths, but good up until 100bp

PhiX % aligned

0.00% PhiX aligned for either run (it was added at 2%)

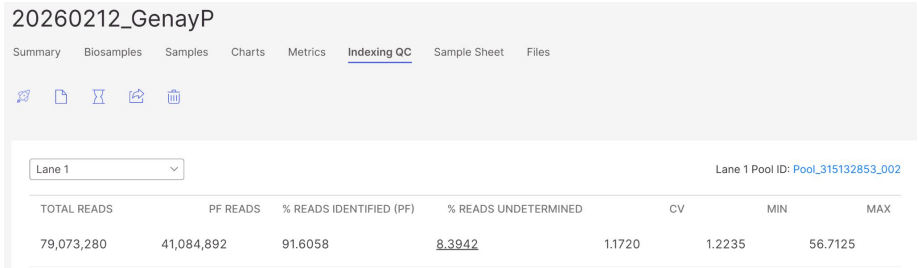
Unfortunately, therefore no error rate was calculated

Hypothesis:

Overamplification -> inaccurate quant -> overloading of MiSeq i100

Demux report: runs are within 1% of each other for successful demux

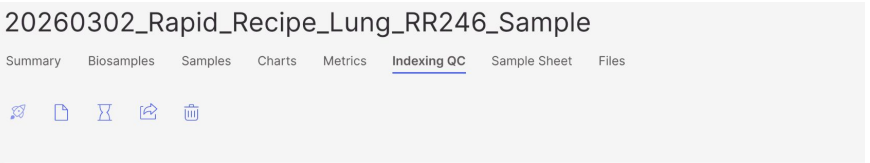
Standard: 91.6% Reads Identified



Per Index

INDEX	SAMPLE ID	LIBRARY NAME	INDEX 1 (I7)	INDEX 2 (I5)	% READS IDENTIFIED (PF)
1	RR246001	n/a	ACATATCCAG	ATTGCACATA	56.7125
2	RR246002	n/a	TCATAGATTG	CACCTTAATC	1.2235
3	RR246003	n/a	GTATTCCACC	TTGTCTACAT	32.2396
4	RR246004	n/a	CCTCCGTCCA	CACCGATGTG	1.4303

Turbo: 90.7% live demux



Live % Demux
90.7

Per sample

Sample ID	Index 1	Index 2	% Demux
RR246001_Lung_Tissue	ACATATCCAG	TATGTGCAAT	56.5275
RR246002_Neg_Ctrl1	TCATAGATTG	GATTAAGGTG	1.2175
RR246003_Plasma	GTATTCCACC	ATGTAGACAA	31.6275
RR246004_Neg_Ctrl2	CCTCCGTCCA	CACATCGGTG	1.3275

Analysis

Questions

- Would we draw the same conclusion (pathogen ID) if we used the Turbo recipe?
- Can we get a good consensus genome from the Turbo recipe?

Analysis groups

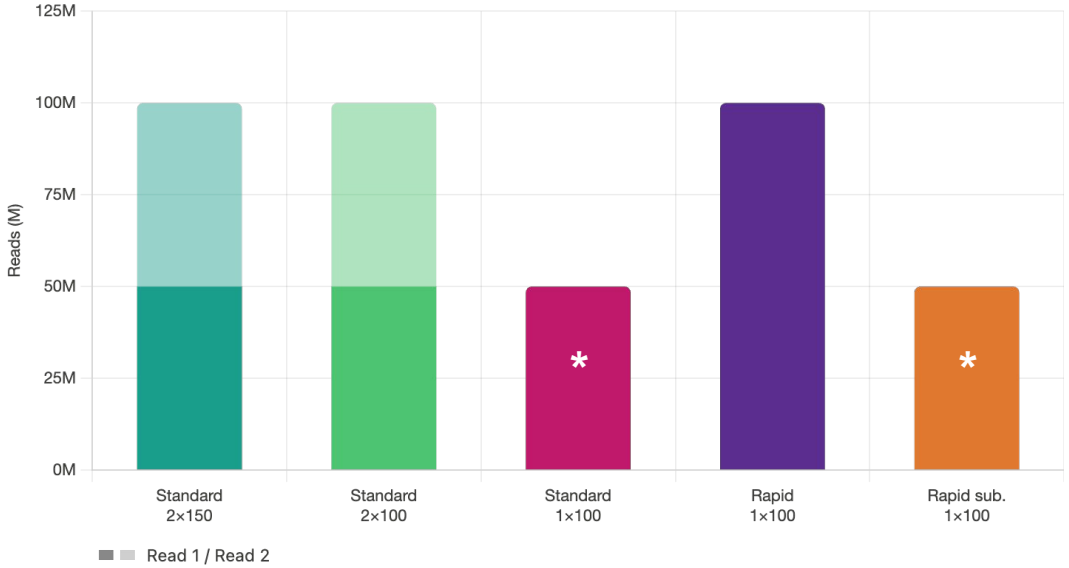
- Turbo 1x100 (100M)
- * Turbo 1x100 (subsampled)
- Standard 2x150 (50M)
- Standard 2x100 (trimmed, 50M)
- * Standard 1x100 (trimmed, 50M)

Run type	Read length	Clusters	Read type
Turbo	1x100	100M	Single
Turbo	1x100	subsampled to match standard 1x100, 50M	Single
Standard	2x150	50M	Paired
Standard	2x100 trimmed	50M	Paired
Standard	1x100 trimmed	50M	Single

Based on the original run conditions, there is no perfect comparison.
** indicates the best comparison, though it takes a hit on number of reads due to subsampling.*

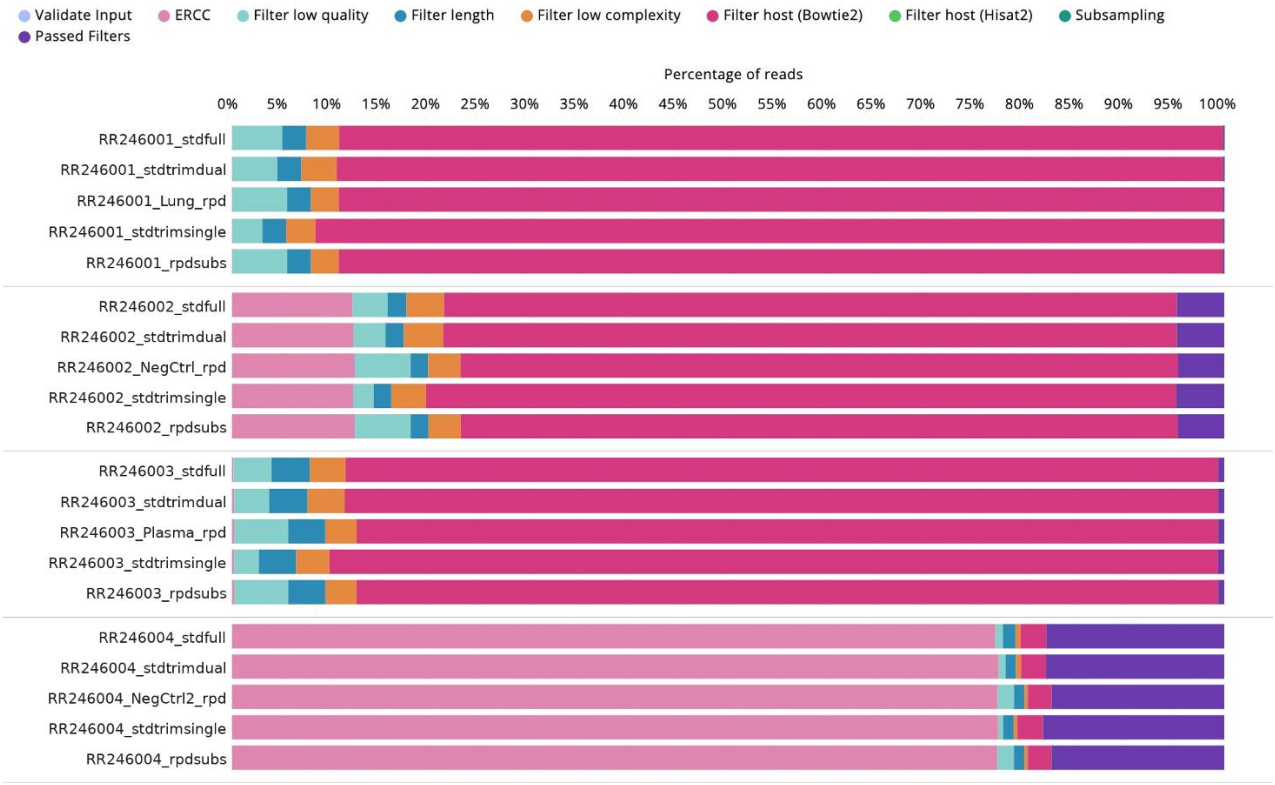
Analysis groups

- Turbo 1x100 (100M)
- * Turbo 1x100 (subsampled)
- Standard 2x150 (50M)
- Standard 2x100 (trimmed, 50M)
- * Standard 1x100 (trimmed, 50M)



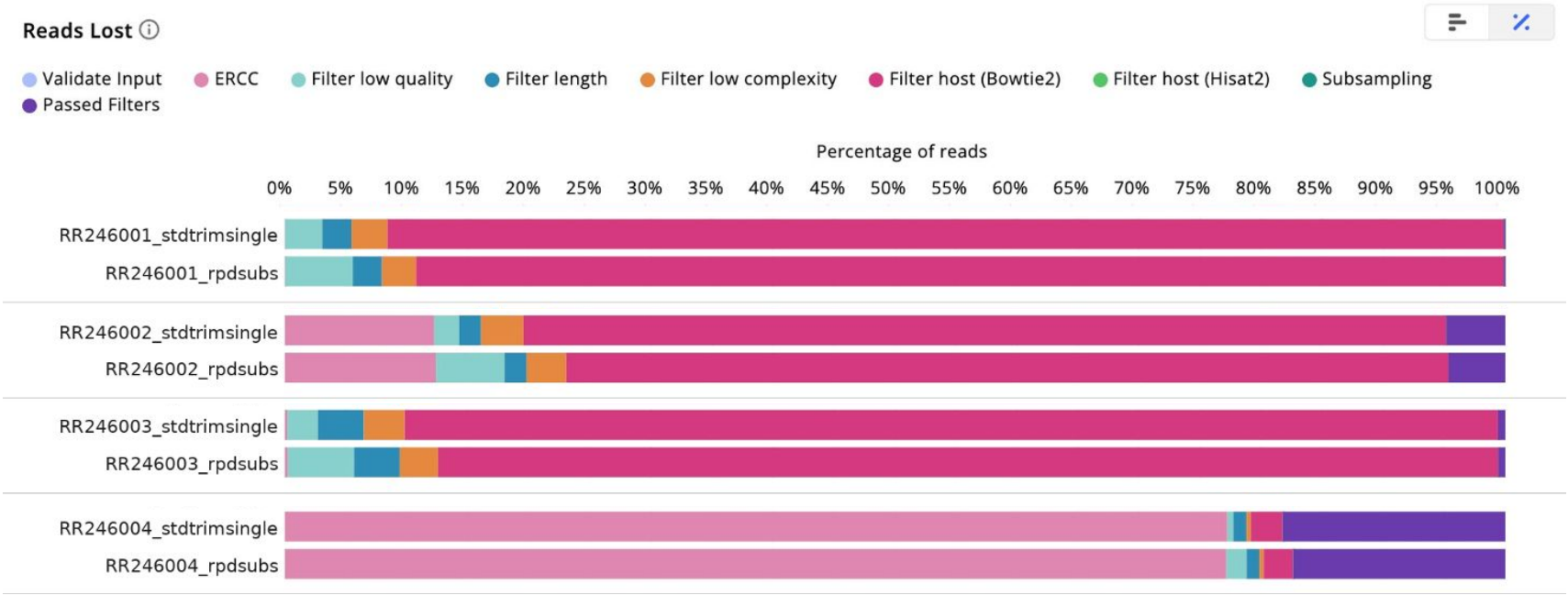
Based on the original run conditions, there is no perfect comparison.
** indicates the best comparison, though it takes a hit on number of reads due to subsampling.*

Reads QC (CZ ID) - all conditions



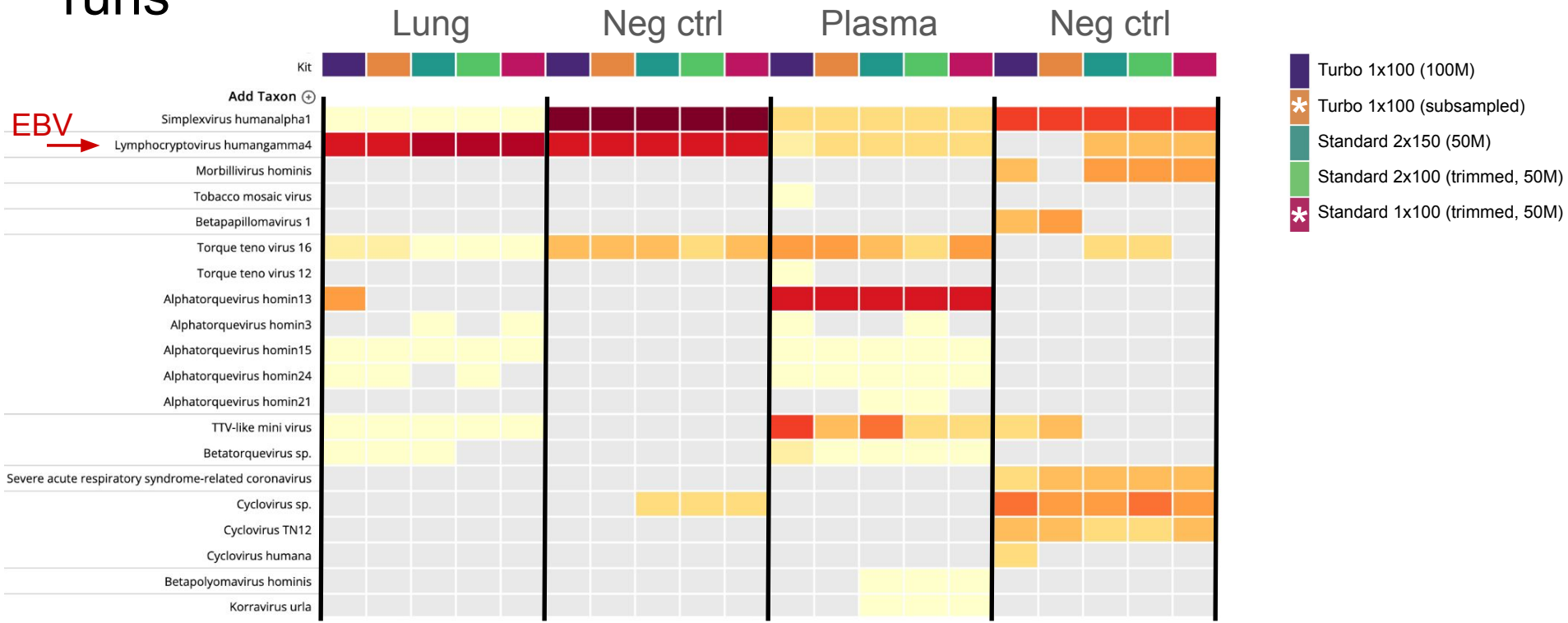
No huge differences; Turbo run has slightly more reads filtered for low quality

Reads QC (CZ ID) - apples to apples comparison



No huge differences; Turbo run has slightly more reads filtered for low quality

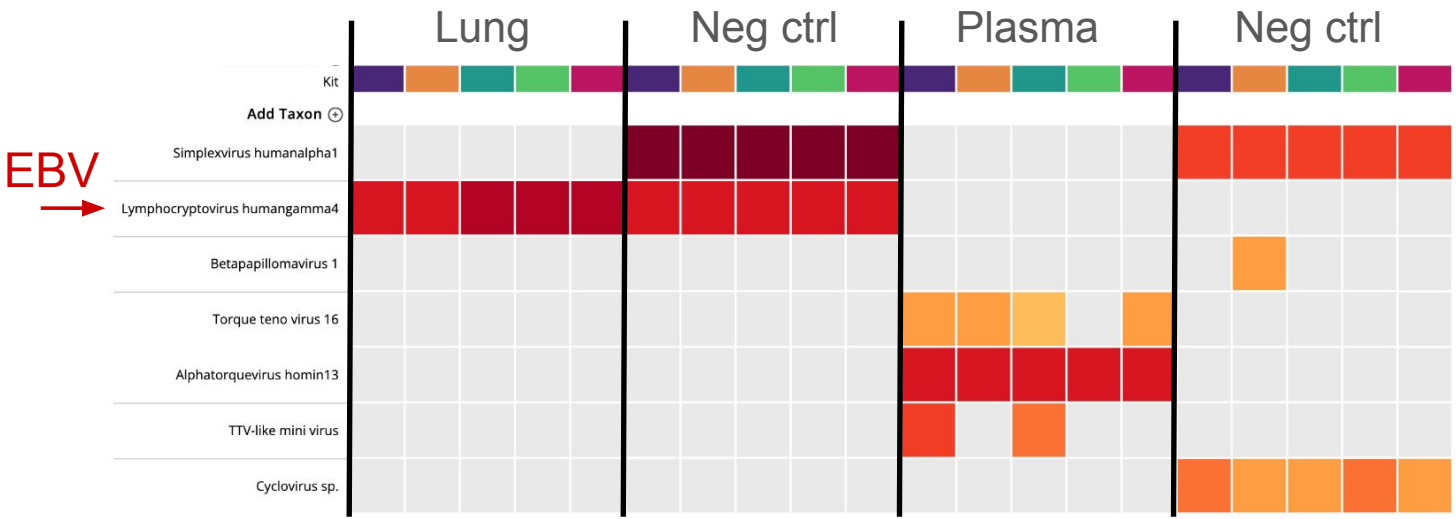
Viruses, **no filters** shows few differences between seq runs



I do not know why the Alhpatorquevirus 13 is showing up in lung Turbo

Displaying NT rPM

Viruses with filters applied show few differences

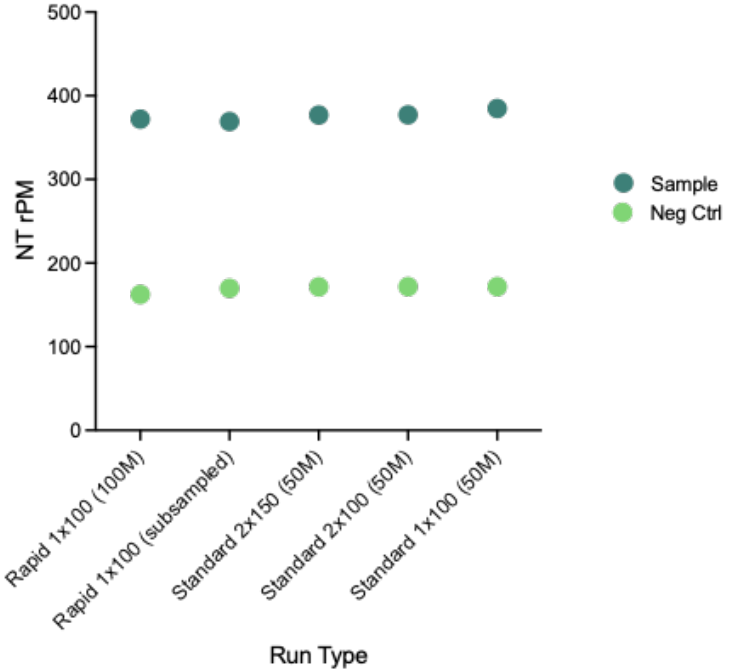


- Turbo 1x100 (100M)
- Turbo 1x100 (subsampled)
- Standard 2x150 (50M)
- Standard 2x100 (trimmed, 50M)
- Standard 1x100 (trimmed, 50M)

Filters:

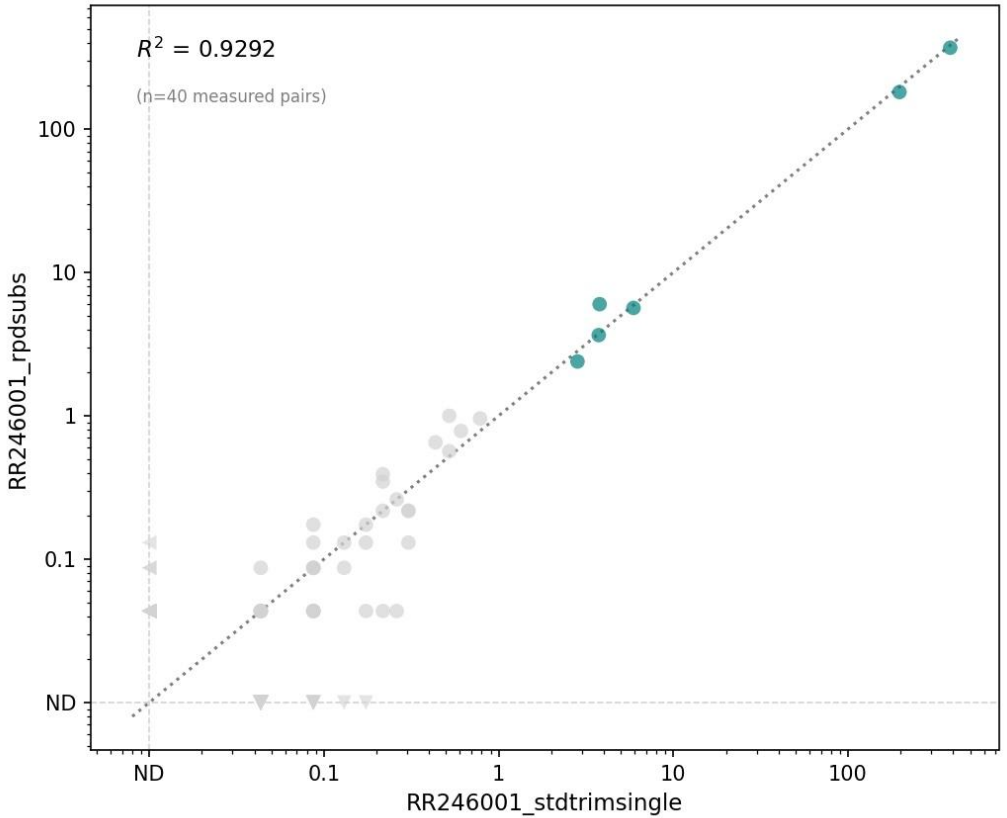
- NT rPM ≥ 10
- NT L ≥ 50
- NR r ≥ 1

EBV rPM is comparable across methods



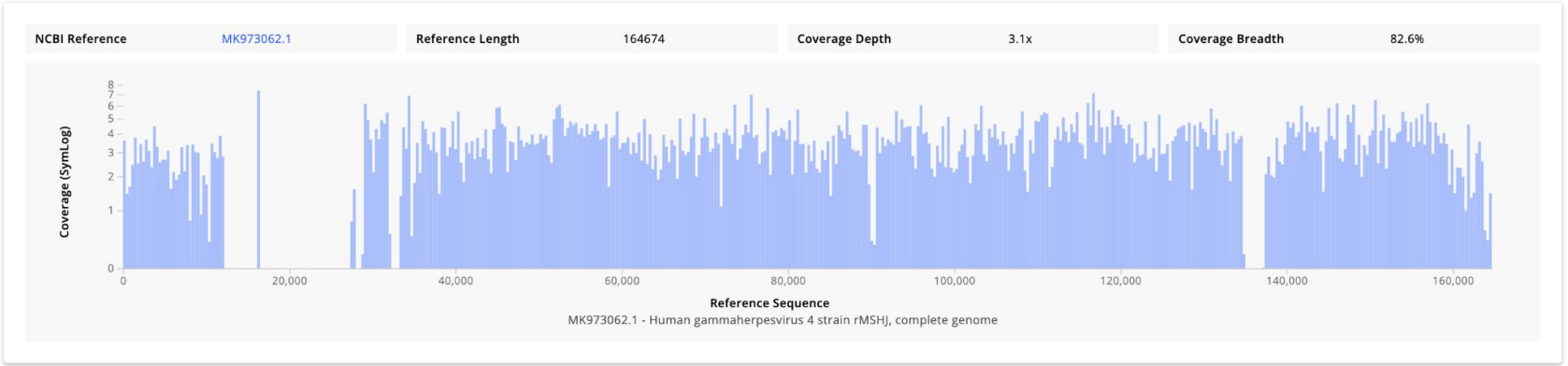
Based on this, the heatmap, and the coverage plots, we would draw the same pathogen detection conclusion with all run conditions

Good correlation among taxa detected in both samples

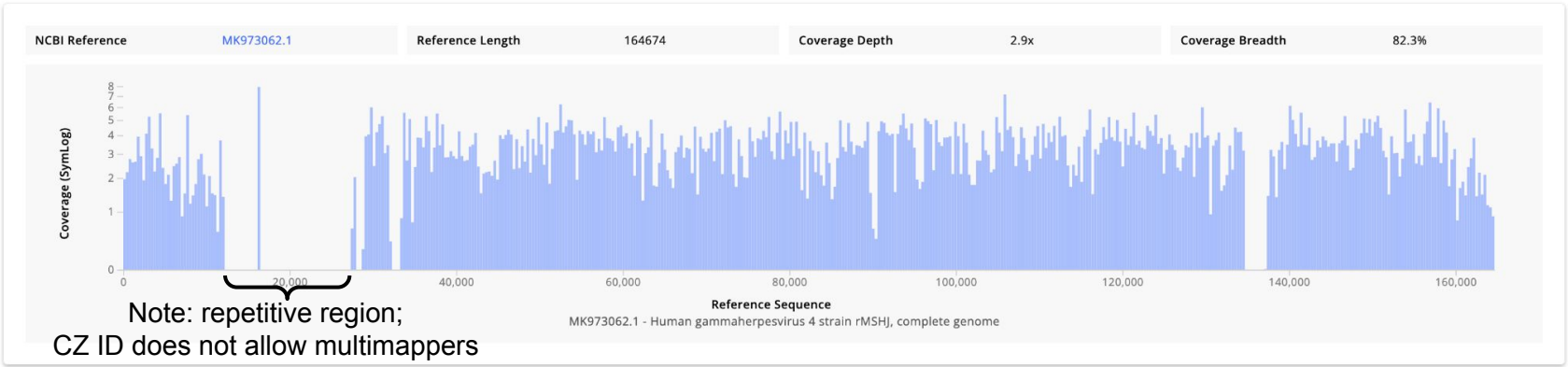


Consensus genome coverage is nearly identical

Standard 1x100 (50M)



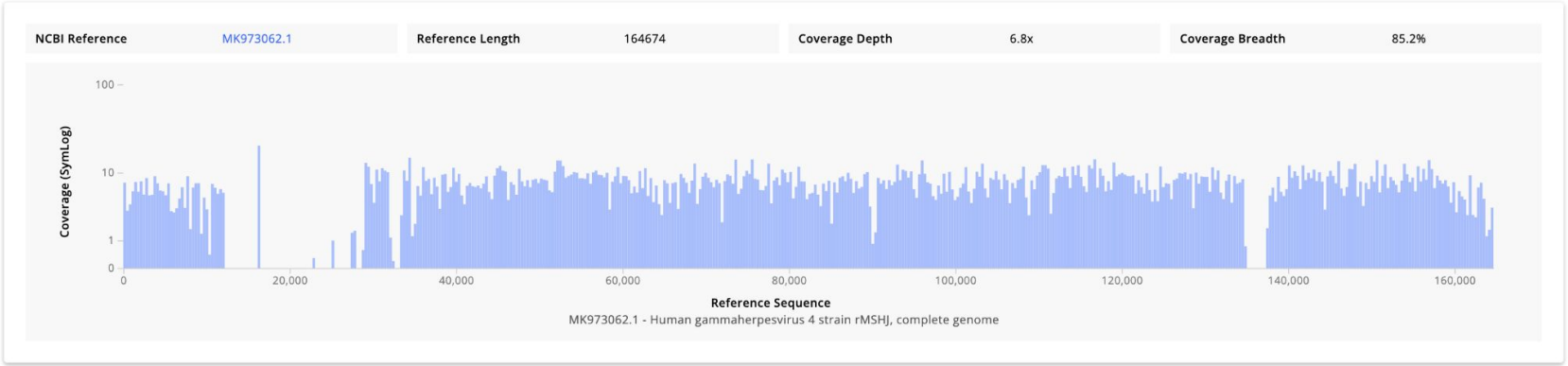
Turbo 1x100 subsampled



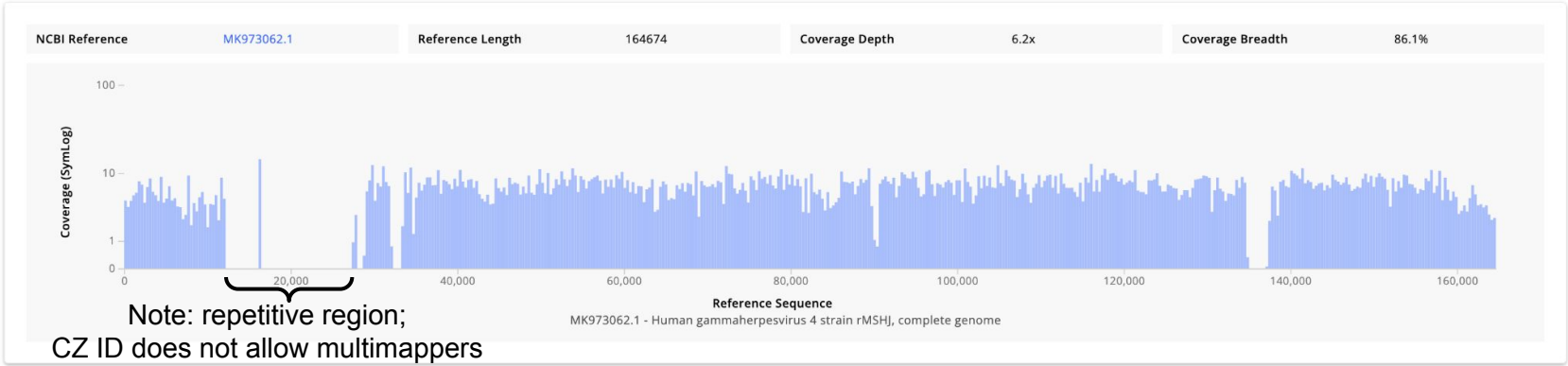
Note: repetitive region;
CZ ID does not allow multimappers

Genome coverage is better with more reads

Standard 2x150 (50M)



Turbo recipe 1x100 (100M)



Note: repetitive region;
CZ ID does not allow multimappers

Consensus genomes are better with more reads, but don't seem to differ notably between run conditions

	Turbo	Standard	Standard trimmed	Turbo subsampled	Standard trimmed R1
Run parameters	1x100 (100M)	2x150 (50M)	2x100 (50M)	1x100 (sub sample)	1x100 (50M)
Coverage depth	6.2x	6.8x	6.1x	2.9x	3.1x
Coverage breadth	86.1%	85.2%	85.1%	82.3%	82.6%
% Genome called (5x depth)	65.9%	66.1%	62.8%	23.1%	25.9%
Ambiguous bases*	542	23	19	145	758
Mapped reads	11916	10201	11563	5696	5959

* Number of degenerate bases (non-C, T, G, or A). Requires 75% frequency to call a base.

(CZ ID consensus stats)

Summary

- Turbo recipe looks good
- Increasing reads (100M kit) can overcome most of the downsides of the 1x100 reads when compared to a dual run on a 50M kit